

**RMO(INDIA) – 1990**

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
  - Ruler and compass allowed.
  - Answer all the questions.
1. Two boxes contain between them 65 balls of different sizes. Each ball is white, black, red or yellow. If you take any 5 balls of the same color at least two of them will always be of the same size. Prove that there are at least 3 balls which lie in the same box have the same color and have the same size.
  2. For all positive real numbers  $a, b, c$  prove that  $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$ .
  3. A square sheet of paper ABCD is so folded that B falls on the mid point M of CD. Prove that the crease will divide BC in the ratio 5:3.
  4. Find the remainder when  $2^{1990}$  is divided by 1990.
  5. P is any point inside a triangle ABC. The perimeter of the triangle  $AB + BC + CA = 2s$ . Prove that  $s < AP + BP + CP < 2s$ .
  6. N is a 50 digit number ( in decimal scale). All digits except the 26<sup>th</sup> digit (from the left) are 1. If N is divisible by 13, find the 26<sup>th</sup> digit.
  7. A census man on duty visited a house which the lady inmate declined to reveal their individual ages, but said – “we do not mind giving you the sum of ages of any two ladies you may choose”. There upon the census man said – “In that case please give me the sum of the ages of every possible pair of you”. They gave the sum as 30, 33, 41, 58, 66, 69. The census man took these figures and happily went away. How did he calculate the individual ages of the ladies from these figures?
  8. If the circum centre and centroid of a triangle coincide, prove that the triangle must be equilateral.

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